Journal of Clinical Epidemiology 158 (2023) 70−83

Journal of Clinical Epidemiology

GRADE GUIDANCE SERIES

GRADE guidance 36: updates to GRADE's approach to addressing inconsistency

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Accepted 1 March 2023; Published online 9 March 2023

Abstract

Objectives: To update previous Grading of Recommendations Assessment, Development and Evaluation (GRADE) guidance by addressing inconsistencies and interpreting subgroup analyses.

Study Design and Setting: Using an iterative process, we consulted with members of the GRADE working group through multiple rounds of written feedback and discussions at GRADE working group meetings.

Results: The guidance complements previous guidance with clarification in two areas: (1) assessing inconsistency and (2) assessing the credibility of possible effect modifiers that might explain inconsistency. Specifically, the guidance clarifies that inconsistency refers to variability in results, not in study characteristics; that inconsistency assessment for binary outcomes requires consideration of both relative and absolute effects; how to decide between narrower and broader questions in systematic reviews and guidelines; that, with the same evidence, ratings of inconsistency may differ depending on the target of certainty rating; and how GRADE inconsistency ratings relate to a statistical measure of inconsistency I^2 depending on the context in which one views results. The second part of the guidance illustrates, based on a

Author Contributions: G. G. has contributed to conceptualization, methodology, writing - original draft, writing - review & editing, and project administration. Y. Z. has contributed to conceptualization, methodology, writing - original draft, writing - review, and editing. M. M. has contributed to methodology, writing - review, and editing. M. B. has contributed to methodology, writing - review, and editing. R. M. has contributed to methodology, writing - review, and editing. A. I. has contributed to methodology, writing - review, and editing. M. H. has contributed to methodology, writing - review, and editing. A. I. has contributed to methodology, writing - review, and editing. A. C. A. has contributed to methodology, writing - review, and editing. F. F. has contributed to methodology, writing - review, and editing. X. S. has contributed to methodology, writing - review, and editing. H. S. has contributed to methodology, writing - review, and editing. H. D. has contributed to methodology, writing - review, and editing. E. A. A. has contributed to methodology, writing - review, and editing. R. C. has contributed to methodology, writing - review, and editing. S. S. has contributed to methodology, writing - original draft, writing - review & editing, and project administration.

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https://doi.org/10.1016/j.jclinepi.2023.03.003

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worked example, the use of the instrument to assess the credibility of effect modification analyses. The guidance explains the stepwise process of moving from a subgroup analysis to assessing the credibility of effect modification and, if found credible, to subgroup-specific effect estimates and GRADE certainty ratings.

Conclusion: This updated guidance addresses specific conceptual and practical issues that systematic review authors frequently face when considering the degree of inconsistency in estimates of treatment effects across studies. © 2023 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Keywords: GRADE; Certainty in evidence; Inconsistency; Heterogeneity; Systematic review; Effect modification; Subgroup analysis

1. Introduction

In 2011, Grading of Recommendations Assessment, Development and Evaluation (GRADE) published, as part of its Journal of Clinical Epidemiology series, initial guidance on rating down the certainty of evidence for inconsistency between studies [1]. Further development of relevant methods, particularly with regard to rating the credibility of subgroup analyses, suggested the desirability of an update of GRADE guidance in this area. Here, we provide that updated guidance. We organized the discussion in two sections. The first deals with general issues regarding whether to rate down for inconsistency. The second addresses the credibility of subgroup analyses.

Previous GRADE inconsistency guidance focused exclusively on binary outcomes. This updated guidance, while still focusing primarily on binary outcomes, also addresses continuous outcomes. Specifically, we provide guidance on the use of I^2 in the context of continuous outcomes and note that, when deciding on the credibility of subgroup analyses for continuous outcomes, our focus is typically on absolute rather than relative effects.

2. General issues in rating down for inconsistency

2.1. Inconsistency: study designs (PICO) or study results?

Typically, studies addressing a particular clinical question will vary in their study design. Variations may include differences in eligibility criteria (and thus, to varying extents, the patients enrolled), the intervention (e.g., different doses), the comparator (e.g., differences in standard of care), and outcomes (e.g., duration of follow-up), or the risk of bias. The choice of study question (population, intervention, comparator, outcome − the PICO) is fundamental to the study design. Some might use the term "inconsistency" (or clinical heterogeneity) to characterize variability in the PICO addressed, and if that variability between studies is large, consider rating down the certainty of the evidence for this inconsistency.

GRADE sees this situation very differently. The variability that the GRADE inconsistency domain is addressing is not variability in study design, but rather variability in study results (e.g., treatment effects in systematic reviews of clinical trials or risks in systematic reviews of prognostic studies). Indeed, variability in the study designs of eligible studies in a systematic review represents at the very least an opportunity, and perhaps even a strength, of any systematic review.

First, if results prove similar from study to study, the variability in study PICOs enhances the generalizability of those results. If patient populations differ widely and results are nevertheless similar across studies, it tells us that patients with different characteristics respond similarly to the intervention, allowing application of the results to a wide range of patients. If studies use different doses but results are nevertheless similar across studies, this tells us intervention effects are independent of dose. Clinicians can then choose freely from among the doses used in the studies (e.g., use lower doses that are less likely to cause adverse effects). Similar interpretations would apply to differences in the comparator and outcomes.

Second, if results differ substantially from study to study, differences in study design (i.e., PICOs) offer an opportunity to explore possible subgroup effects. For instance, if studies differ in the populations they enroll (e.g., focus on young vs. old), it offers an opportunity to explore whether effects differ in the subpopulations (greater or lesser effects in the young vs. the old). If studies differ in interventions (e.g., high vs. low dose), it offers an opportunity to explore whether effects differ according to intervention characteristics (greater or lesser effect in high vs. low dose).

Our discussion will explore, in some detail, the issue of dealing with inconsistency in results and how subgroup analysis can help us understand whether differences in studies' PICOS explain inconsistency in results.

2.2. Relative and absolute effects

The original GRADE guidance for inconsistency dealt exclusively with binary outcomes such as death or stroke, and noted that relative effects very seldom show important differences across groups [2−4]. Thus, despite relative effects being similar, in groups that vary appreciably in baseline risks, absolute effects will differ substantially.

Because relative effects tend to be similar across subgroups but absolute effects do not, the GRADE approach to rating down for inconsistency applies, for binary outcomes, to relative effects. For continuous outcomes such as pain or duration of hospitalization, the GRADE approach

What is new?

Key findings

- When assessing inconsistency, Grading of Recommendations Assessment, Development and Evaluation (GRADE) users should consider both relative and absolute effects.
- Systematic reviewers and guidelines developers should generally choose broad questions and, if they suspect they may confront inconsistent results across studies, generate a small number of priori hypotheses regarding possible explanations, including the direction of the hypothesized effect.
- Reviewers need to understand the I^2 statistic and the contexts when I^2 is large but important inconsistencies are not present.
- The instrument to assess the credibility of effect modification analyses (ICEMAN) is fully compatible with GRADE's approach to addressing inconsistency.

What this adds to what is known.

- With the same evidence, ratings of inconsistency may differ depending on the target of certainty rating
- ICEMAN provides a useful instrument to inform GRADE certainty ratings of the credibility of subgroup effects, acknowledging that credibility represents a continuum from high to very low.

What is the implication, and what should change now?

- To address inconsistency optimally, GRADE users should familiarize themselves with the new guidance presented in this article that augments previous guidance.
- When judging the credibility of subgroup analyses, GRADE users should consider using the formal, expert consensus-generated, pretested ICEMAN instrument.

focuses on differences in absolute effects, typically differences in means.

For binary outcomes, however, GRADE addresses how one should deal with differences in baseline risk across patients, differences that will result in variability in absolute effects. Review authors and guideline developers should construct a priori hypotheses addressing heterogeneity. Ideally, for binary outcomes, they would construct such hypotheses for both relative and absolute effects. We expand on these issues in the next sections of this article.

The issue of relative effects differing through effect modification (the context in which we usually use the term "subgroup effect") as opposed to variation in absolute effects due to differences in baseline risk, raises an issue of terminology. Whether the distinction becomes clear by specifying subgroup effects for relative and absolute (as in Table 1, and as we will do in the rest of this discussion), or by restricting the term "subgroup effect" to relative effects, remains uncertain.

2.3. Choice of the question of interest: broader or narrower

Since meta-analyses generate single estimates of effect, they assume that these estimates apply, more or less, across all relevant patients and interventions covered by the review's PICO. Typically, systematic reviews will conduct their analyses addressing relative effects and infer that the single pooled estimates of relative effects are widely applicable. If that is not the case – if, for instance, older and younger patients or higher and lower doses of medication, differ markedly in intervention effects – results may be profoundly misleading.

This creates a potentially difficult choice for the systematic review author, or the guideline developer. Choose a narrow question (i.e., a narrow range of PICO) and likely sacrifice both precision and generalizability or choose a broader question and risk misleading results.

The solution is to differentiate between three situations, where one is: 1) confident that the effects differ across the patient or intervention factors of interest, 2) confident that they do not differ, or 3) unsure. As the already-introduced Table 1 demonstrates (and as we will expand upon in the next section), when considering binary outcomes, hypotheses regarding differences in effects across patient groups may include relative effect modification and, through differences in baseline risk, varying absolute effects.

Consider, for instance, older and younger patients. The three situations and the actions they dictate would be as follows:

A) One is confident, on the basis of prior evidence that effects differ between older and younger patients. One would then construct two different PICOs and two different recommendations: one for older and one for younger patients.
B) One has no compelling reason to suspect different effects in younger and older patients, and, even if the effects might differ, there is no biological rationale that would lead one to think that effects are larger in the older or younger patients. One would then have a single PICO, with results applying to both groups of patients.

C) One is unsure whether effects are similar in older and younger patients. One would then have a broad single PICO but prespecify age as an effect modifier or a determinant of baseline risk. For relative effects, one would hypothesize different effects in the two subgroups, specify the direction of any effect modification (i.e., larger in older or larger in younger), and subsequently statistically test for effect modification. For absolute effects, one would look for major differences in baseline risk between old and young (for almost any bad event, higher in the older) and, if differences are sufficiently large, generate different estimates of the absolute effect in the two groups.

2.4. Addressing heterogeneity in effects

Let us assume we are in the third situation above: we are uncertain regarding the modification of relative effects or differences in baseline risk. Consider, for instance, the framing of a PICO with associated hypotheses from a recent World Health Organization recommendation for nirmatrelvir/ritonavir in patients with nonsevere coronavirus disease 2019 (COVID-19) to prevent hospitalization from COVID-19 (Table 1) [5]. For relative effects, the panel postulated four possible effect modifiers: age (effects greater in older), time from symptom onset (effects greater with shorter time), serological status (effects greater in those without antibodies), and disease severity (effects greater in less severe). For absolute effects, the panel suspected that baseline risk (and thus absolute effects) would differ across risk groups, serological status, and vaccination status.

If one finds credible and important differences in hypothesized relative effects, GRADE suggests summarizing these differences in separate summary of findings tables and, in the guideline context, providing different recommendations for the different subgroups. On the other hand, if one identifies patient characteristics associated with different baseline risks – ideally from observational studies addressing prognosis – these can appear together in a single summary of findings table and may or may not warrant different recommendations for different risk groups.

For instance, in a World Health Organization recommendation addressing Ebola virus disease [6], the guideline team noted that the risk of dying differed markedly across epidemics in different settings, and therefore chose to present absolute risks for both the highest and lowest risk settings (Table 2). The panel ultimately made the same recommendations (in this case, a strong recommendation in favor of Mab114, and an antibody treatment) for both settings.

Table 1. WHO Framing of PICO (population, intervention, comparator, outcome) question for nirmatrelvir/ritonavir for nonsevere COVID-19 patients including subgroups for both relative and absolute effects (in: BMJ. 2020 Sep 4; 370:m3379)

Nirmatrelvir/ritonavir PICO question	
Population	Patients with nonsevere confirmed COVID-19 (according to WHO definition)
Intervention	Nirmatrelvir/ritonavir + usual care
Comparator	Usual care
Outcome	Hospitalization
Subgroups for relative effects	Candidate effect modifiers 1. Children vs. adults vs. older people 2. Time from symptom onset 3. Serological status 4. Disease severity
Subgroups for absolute effects	Likely determinants of baseline risk 1. High vs. low risk of hospitalization 2. Serological status 3. Vaccination status

2.5. The target of the certainty rating

In the original guidance, GRADE identified four issues to consider in addressing consistency: similarity of point estimates, overlap of confidence intervals, the statistical test of heterogeneity, and the I^2. The guidance also included an assessment of the importance of the differences in study results across subgroups. The four essential criteria remain largely unchanged (with additional limitations of the latter two statistical criteria – see following), and the issue of importance remains germane with a fundamental additional conceptual advance.

In 2017, GRADE established that we are rating certainty in whether effects lie on one side of a threshold or in a particular range [7]. Establishing the target – that is, the particular threshold or range – requires consideration of what we have called the context of the rating. The possible thresholds are the null, the minimally important difference, and the net benefit (the threshold of recommending for or against an intervention); the possible ranges trivial, small, moderate and large effects [7,8].

The choice of the target of a certainty rating has implications for rating down for inconsistency. Results may differ from study to study, but if all point estimates lie above the chosen threshold, rating it down is likely inappropriate. On the other hand, with exactly the same degree of inconsistency, if point estimates lie substantially on either side of the threshold, rating down for inconsistency, is likely appropriate.

For instance, consider a systematic review addressing postoperative pain treatment after total knee arthroplasty [9]. In a meta-analysis of 13 randomized trials, the authors reported that local infiltration analgesia resulted in a mean reduction in pain at 6 hours of 14.14 points on a 100-point scale, with an associated P-value of <0.00001 and an I^2 of 95% (Fig. 1) [9]. Given that 12 of the 13 estimates favored treatment and the 13th showed no difference, and the low likelihood of seeing results like these if the null hypothesis

Table 2. Mab114 vs. standard of care in patients with Ebola virus disease considering two patient groups at different baseline risks (in: Lancet Microbe. 2022 Sep; 3(9):e683-e692)

Outcome	Study results and measurements	Absolute effect estimates		Certainty of the evidence (quality of evidence)	Plain language summary
		Standard care	MAb114		
Mortality (absolute effect estimated from lowest baseline risk)	Relative risk: 0.42 (CI 96% 0.19 – 0.93) based on indirect evidence	**395** per 1,000	**166** per 1,000	**Moderate certainty** Due to serious imprecision	**MAb114 probably reduces mortality compared with standard care when using the lowest baseline risk estimate.**
		Difference: **229 fewer per 1,000** (CI 95% 320 fewer–28 fewer)			
Mortality (absolute effect estimated from highest baseline risk)	Relative risk: 0.42 (CI 96% 0.19 – 0.93) based on indirect evidence	**660** per 1,000	**277** per 1,000	**Moderate certainty** Due to serious imprecision	**MAb114 probably reduces mortality compared with standard care when using the highest baseline risk estimate.**
		Difference: **383 fewer per 1,000** (CI 95% 535 fewer–46 fewer)			
Serious adverse events	Risk difference: 0.016 (CI 95% −0.061 – 0.093) based on indirect evidence	Difference: **16 more per 1,000** (CI 95% 61 fewer - 93 more)		**Very low certainty** Due to serious risk of bias, due to very serious imprecision	**Whether MAb114 increases serious adverse events compared with standard care is very uncertain.**
Time to viral clearance	Measured by: day lower better Based on indirect evidence	**8.86** Mean	**7.54** Mean	**Low certainty** Due to very serious imprecision	**MAb114 may have little or no effect on time to viral clearance compared with standard care.**
		Difference: **MD 1.14 lower** (CI 95% 4.09 lower - 1.81 higher)			

were true, were one to rate one's certainty in a nonzero effect, the decision not to rate down for inconsistency would be secure (illustrating the limitations of I^2 to inform inconsistency decisions, especially when pooling continuous outcomes or absolute effects, a point we will address in the next section).

Consider if, however, one were rating certainty on a minimally important difference (MID) set at 10 points (a frequently chosen threshold in a 100-point pain visual analog scale). Now, the results of 6 of 13 studies fall at or below the MID threshold, with the upper boundary of two studies excluding an important difference. This degree of inconsistency raises doubts about whether local infiltration analgesia achieves an important benefit, and likely warrants rating down for inconsistency.

2.6. I^2 requires contextualization

Previous GRADE guidance noted limitations in I^2 [1]. In this section, we address, in addition, I^2 limitations in the setting of reviews, including larger studies, and assessing continuous outcomes.

As noted in the previous guidance, on the one hand, small studies may have extremely variable point estimates, but with widely overlapping confidence intervals and a low I^2. On the other hand, very large studies may present a consistent clinical message, but they may have narrower confidence intervals with little overlap and a high I^2 [10].

What explains the above behavior? I^2 represents not an absolute but rather a relative measure of statistical heterogeneity. It is a sort of signal-to-noise ratio that provides

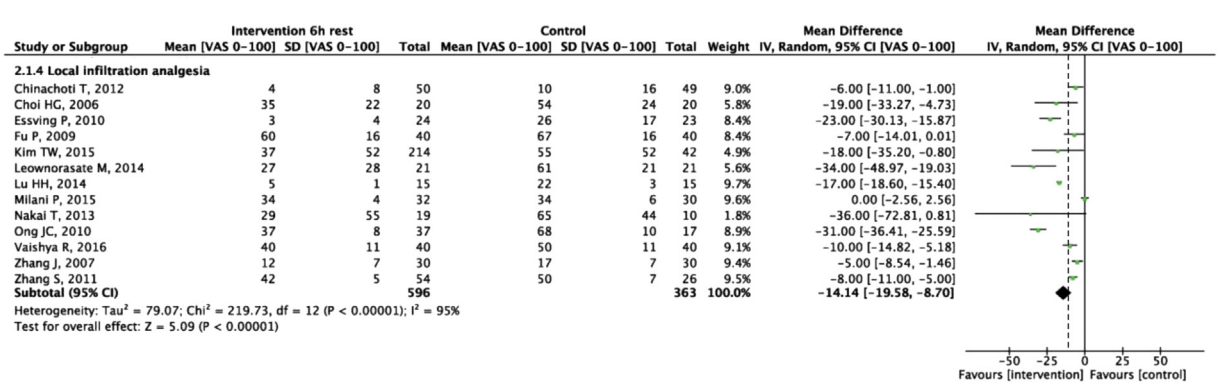


Fig. 1. Impact of local infiltration analgesia on postoperative pain in patients after total knee arthroplasty (in: PLoS One. 2017; 12(3):e0173107). Broken vertical line represents minimal important difference.

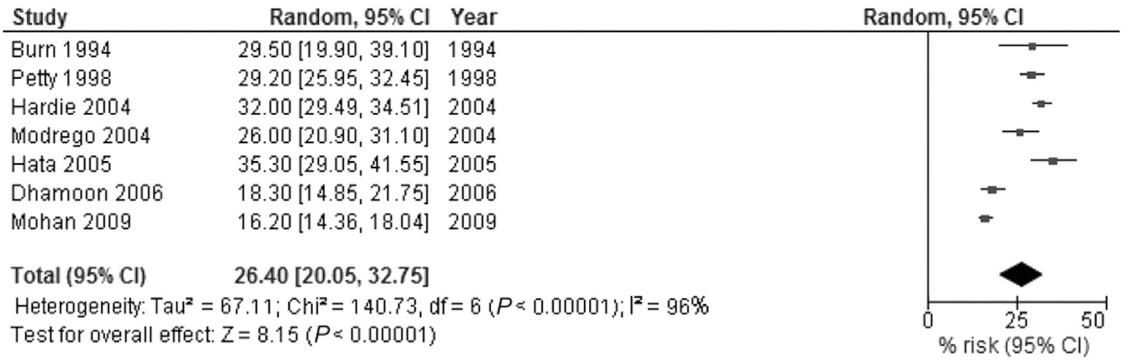


Fig. 2. Risk of stroke recurrence (in: stroke. 2011; 42(5):1489–1494).

insight into the question: Of the variability among estimates that we see in these data, what proportion is attributable to genuine differences as opposed to variability attributable to sampling error (or chance)? [11,12].

GRADE publications introducing ratings of certainty for prognostic studies have emphasized how narrow confidence intervals can result in a high I^2 [13,14], though this is not restricted to studies of prognosis. The large sample sizes often available in prognostic observational studies – and occasionally in therapy observational studies and less commonly in randomized trials – may result in very narrow confidence intervals. If that is the case, one may find point estimates that are similar but, due to minimal overlap in the confidence intervals, those same data may still yield a high I^2. The high I^2 just suggests that the variability – in this case, inconsequential – is mostly real and not due to sampling error, not that the variability impacts the inferences one might draw from the evidence.

If, instead, we had a high I^2 but point estimates that varied substantially, this suggests the substantial variability is mostly real and not due to sampling error, and thus that the evidence is limited by substantial inconsistency. In both cases, the high I^2 contributed to the "mostly real and not due to sampling error" portion of the inference. In the setting of a high I^2, if there is substantial variation in effects across studies, the real variability may be important, but if results are similar, one can conclude the real variability is unimportant and rating down for inconsistency is inappropriate.

Because of what I^2 does and does not measure, and because it must be interpreted in context, popular thresholds for I^2, such as 25%, 50%, and 75%, to indicate small, moderate, and large degrees of heterogeneity are inappropriate [11,12]. Indeed, in the context of large prognostic studies in which consistent results may nevertheless be associated with a high I^2, GRADE guidance suggests ignoring statistical criteria in favor of visual inspection of similarity in point estimates.

Consider, for instance, a study examining the risk of stroke recurrence (Fig. 2) [15]. All seven point estimates are within 10% of the pooled point estimate of 26.4% and four are within 5% of the point estimate. Despite this, the I^2 is 96%. An inappropriate interpretation of I^2 could cause one, without a moment's hesitation, to incorrectly rate it down for inconsistency. Examining the actual variability in results, if one considered values within 10% of the point estimate of 26.4% to support similar decisions in patient care, one probably would not rate down.

Meta-analyses focusing on continuous outcome variables represent a less dramatic situation that nevertheless requires the same considerations for proper contextualization of I^2 interpretation. Because trials addressing treatment impact on continuous outcome variables can generate

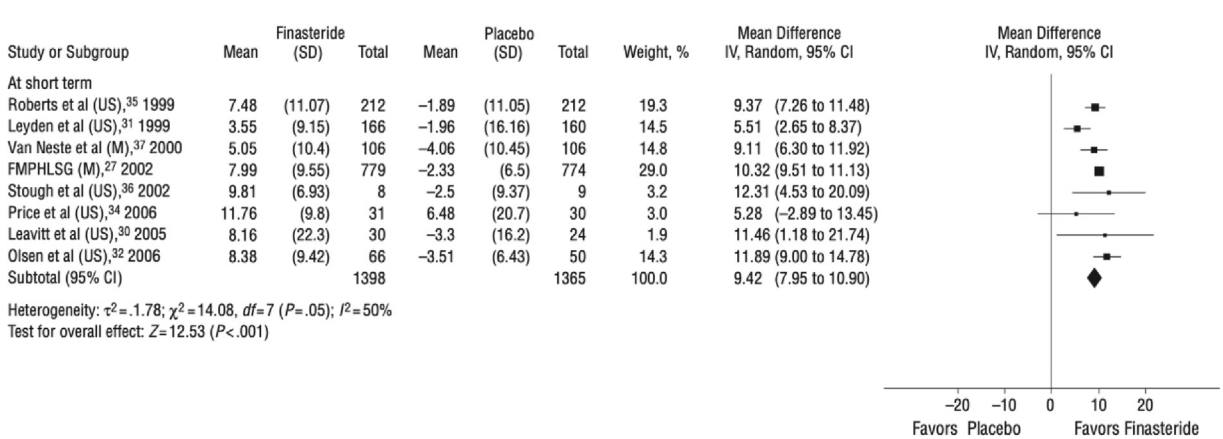


Fig. 3. Forest plot from a systematic review of finasteride in androgenic alopecia addressing hair count (in: Arch Dermatol. 2010; 146 (10):1141–1150).

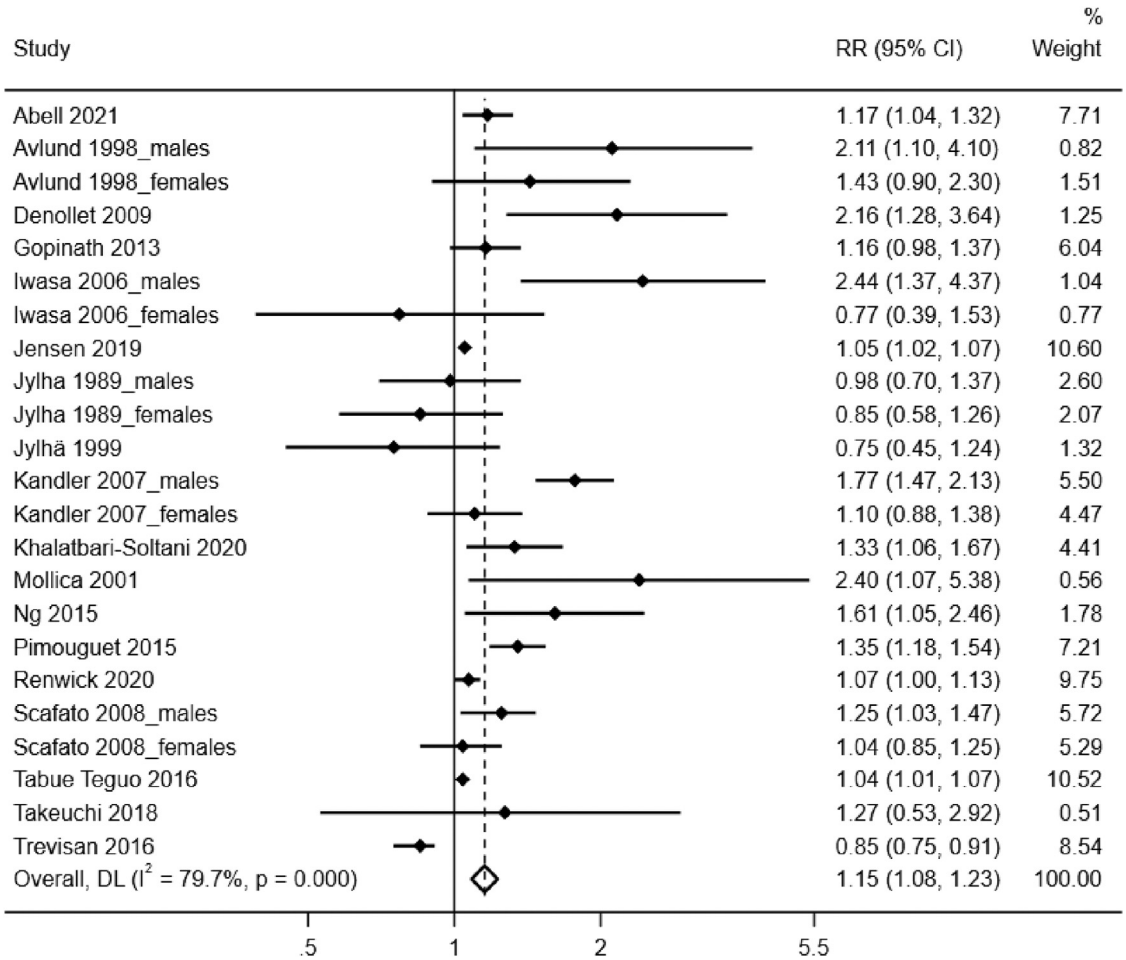


Fig. 4. Forest plot of 23 observational studies addressing the association between living alone and mortality (in: EClinicalMedicine. 2022 Sep 29; 54:101,677).

narrower confidence intervals than those focusing on binary outcomes, a high I^2 is again more likely even if results are similar from study to study. Another reason why I^2, on average, is larger for continuous outcomes is that the typical absolute effect measure, the difference in means, is more susceptible to baseline variation than the relative effect measures used for binary outcomes [16]. One survey addressing this issue reported that 34% of 352 meta-analyses evaluating treatment effects on continuous variables reported an I^2 of 0%, while 52% of 557 meta-analyses evaluating treatment effects on binary variables reported an I^2 of 0. I^2 of 60 to 100% occurred in 39% of continuous and 14% of binary outcomes [17]. Again, over-reliance on I^2 could mislead: the interpretation of I^2 as part of assessing consistency must consider the context.

Consider another example: A systematic review of finasteride for androgenetic alopecia [18]. Figure 3 presents a forest plot of mean differences in hair count in intervention vs. control groups. Here, despite relatively modest sample sizes, the continuous variable outcome generates narrow confidence intervals, contributing to the I^2 of 50%. Looking at the I^2 of 50% in isolation, and misinterpreting it as an absolute measure of heterogeneity, would raise concerns regarding inconsistency. However, seven of nine estimates lie between 9 and 12, and the other two are greater than five, all favoring finasteride, engendering an appropriate conclusion of minimal inconsistency.

Figure 1, described in the previous section, provides another vivid illustration of the need for contextualization. The very high I^2 of 95% is a function of the narrow confidence intervals associated with the measurement of the continuous variable of pain. The high I^2 would not warrant rating down for inconsistency when rating certainty in a nonzero effect. However, if rating certainty that the truth lies above the MID, the I^2 might support a decision to rate it down for inconsistency.

3. Issues specific to subgroup analyses

3.1. Assessing the credibility of subgroup effect

The initial GRADE article on rating down for inconsistency provided detailed guidance for assessing the credibility of postulated subgroup effects. The guidance suggested systematic review authors should always be prepared to find inconsistency (heterogeneity) in results based

on differences in study design and prepare a priori hypotheses that may explain this heterogeneity. Irrespective of the degree of heterogeneity they find, they should test these hypotheses. If they observe substantial heterogeneity that their hypotheses fail to explain, they will rate it down for inconsistency. The remainder of this article will deal with situations in which results raise the possibility that these a priori hypotheses do explain the observed inconsistency between study results.

Subsequent to the original publication addressing GRADE's approach to inconsistency, we have conducted a systematic survey of method papers addressing the credibility of subgroup analyses. The survey identified 30 checklists (including one appearing in the previous GRADE inconsistency guidance) suggesting criteria for making credibility judgments [19]. None of these checklists benefited from a systematic survey of the relevant literature; brought together a panel of experts who have written key articles regarding subgroup analysis methods to thrash out optimal criteria; created a formal instrument incorporating the judgments of the panel; extensively pretested the instrument with trial investigators, systematic review authors, and journal editors; produced a supplementary document providing detailed guidance for using the formal instrument; or produced an online fillable form to complete the instrument. A group of investigators conducted all these steps to produce an assessment tool that addresses the credibility of subgroup analysis − whether a true subgroup effect exists, not whether its magnitude is important − that they call the ICEMAN [20]. While ICEMAN provides separate criteria for individual randomized trials and for meta-analyses, since GRADE addresses bodies of evidence rather than individual studies, the latter is particularly relevant to GRADE. ICEMAN evaluates claims of subgroup effects; a claim of no subgroup effect would require a different instrument. Further, ICEMAN addresses whether an effect modification exists and not its importance.

GRADE does not recommend specific instruments for assessing the risk of bias, applicability, or, in this case, credibility of subgroup analysis. Here, we address the use of ICEMAN, illustrating its use in a recent systematic review that well highlights issues in its application (box). Users of GRADE faced with a possible subgroup effect may choose to use one of the previously available checklists [21].

3.2. The example review; credibility of the subgroup effect; and nine criteria to consider

Although the developers of ICEMAN focused on issues in randomized trials, it turns out − as we shall see − that criteria apply equally well to observational studies. A team of investigators conducted a systematic review and meta-analysis of observational studies addressing the possible association or causal relationship between living alone and mortality [22]. For their GRADE rating of certainty, investigators chose the null effect as the threshold. Figure 4 presents the forest plot summarizing the 23 observational studies − we refer to the Jylha 1989 study in males and females as separate studies − that, albeit with substantial heterogeneity, suggested a 15% increase in the relative risk (RR) of death if living alone.

The authors suspected, however, that the magnitude of the effect might differ in older and younger individuals and set about exploring this hypothesis. We will now apply the ICEMAN criteria to the possible difference in effects between younger and older age groups.

3.3. Applying ICEMAN

Criterion 1: Is the analysis of effect modification based on comparison within rather than between trials? (Box).

Because between-trial effects may differ on a whole host of characteristics other than the putative effect modifier, within-trial comparisons are far more compelling than between-trial comparisons [19]. In addressing the possible subgroup effect, the review therefore included only studies that separately reported on the association between living alone and mortality for younger and older adults. They found five relevant studies − four of which used the threshold of 75 and one of 65, ultimately choosing to include all five studies − considering the two groups as "older" or "younger". Their focus exclusively on within-study comparisons mandates a "completely within" response to the first ICEMAN criterion. Through adjustment for several patient characteristics, and modeling of continuous outcomes, an individual participant data meta-analysis can strengthen inferences beyond even what is possible from a within-trial comparison based on group-level data.

Criteria 2: If two or more within-trial comparisons are available, is the effect modification similar from trial to trial?

In each of the five paired comparisons, the increase in mortality associated with living alone proved greater in the younger individuals than the older individuals, with a ratio of relative risk of 1.1 to 3.22 (Fig. 5). That the ratio of relative risks is always in the same direction suggests a response of "definitely similar", the variability in the ratio suggests a response of "mostly similar". Review authors chose the former.

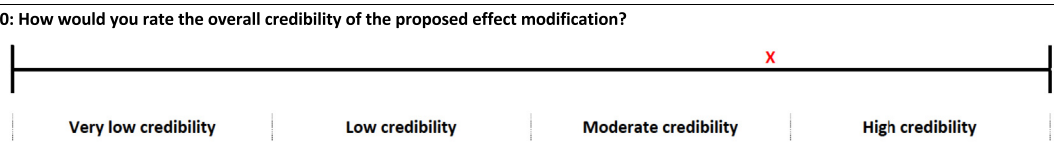
Criteria 3: For between-RCT comparisons, is the number of studies large?

As the subgroup exploration focused exclusively within-trials to conduct comparisons, this criterion was not applicable.

Criteria 4: Was the direction of effect modification correctly hypothesized a priori?

The review protocol did not postulate this particular subgroup effect and thus did not speculate on its possible direction, mandating a response of "definitely no". Some

1: Is the analysis of effect modification based on comparison within rather than between trials?			
[ ] Completely between	[ ] Mostly between or unclear	[ ] Mostly within	[X] Completely within
Subgroup analysis or meta-regression comparing overall effects of each individual trial. This is typical for aggregate data meta-analysis.	*Subgroup analysis or meta-regression with most information coming from overall effects, but some trials providing within-trial subgroup information*	*Most trials providing within-trial subgroup information; or individual participant data analysis that combines within and between trial information*	*All trials providing within-trial subgroup information or individual participant data; and the analysis separates within from between trial information, e.g., meta-analysis of interactions*
2: For within-trial comparisons, is the effect modification similar from trial to trial? [ ] Not applicable: no or one within-RCT comparison			
[ ] Definitely not similar	[ ] Probably not similar or unclear	[ ] Mostly similar	[X] Definitely similar
Effect modification reported for two or more trials and clearly different directions	*Effect modification not reported for individual trials or too imprecise to tell*	*Effect modification reported for two or more trials, mostly similar in direction, but considerable differences in magnitude*	*Effect modification reported for two or more trials, similar in direction, only some differences in magnitude*
3: For between-trial comparisons, is the number of trials large? [X] Not applicable: no between RCT comparison			
[ ] Very small	[ ] Rather small or unclear	[ ] Rather large	[ ] Large
1 or 2 or in smallest subgroup; 5 or less in continuous meta-regression	*3-4 in smallest subgroup; 6-10 in continuous meta-regression*	*5-9 in smallest subgroup; 11 to 15 in continuous meta-regression*	*10 or more in smallest subgroup; more than 15 in continuous meta-regression*
4: Was the direction of effect modification correctly hypothesized a priori?			
[X] Definitely no	[ ] Probably no or unclear	[ ] Probably yes	[ ] Definitely yes
Clearly post-hoc or results inconsistent with hypothesized direction or biologically very implausible	*Vague hypothesis or hypothesized direction unclear*	*No prior protocol available but unequivocal statement of a priori hypothesis with correct direction of effect modification*	*Prior protocol available and includes correct specification of direction of effect modification, e.g., based on a biologic rationale*
5: Does a test for interaction suggest that chance is an unlikely explanation of the apparent effect modification? (consider irrespective of number of effect modifiers)			
[ ] Chance a very likely explanation	[ ] Chance a likely explanation or unclear	[ ] Chance may not explain	[X] Chance an unlikely explanation
Interaction or meta-regression p-value >0.05	*Interaction or meta-regression p-value ≤0.05 and >0.01, or no test of interaction reported and not computable*	*Interaction or meta-regression p-value ≤0.01 and >0.005*	*Interaction or meta-regression p-value ≤0.005*
6: Did the authors test only a small number of effect modifiers or consider the number in their statistical analysis?			
[ ] Definitely no	[X] Probably no or unclear	[ ] Probably yes	[ ] Definitely yes
Explicitly exploratory analysis or large number of effect modifiers tested (e.g., greater than 10) and multiplicity not considered in analysis	*No mention of number or 4-10 effect modifiers tested and number not considered in analysis*	*No protocol available but unequivocal statement of 3 or fewer effect modifiers tested*	*Protocol available and 3 or fewer effect modifiers tested or number considered in analysis*
7: Did the authors use a random effects model? [ ] Not applicable			
[ ] Definitely no	[ ] Probably no or unclear	[ ] Probably yes	[X] Definitely yes
Fixed (or common) effect or fixed effects model explicitly stated	*Probably fixed effect(s) model*	*Probably random (or mixed) effects*	*Random (or mixed) effects explicitly stated*
8: If the effect modifier is a continuous variable, were arbitrary cut points avoided? [ ] not applicable: not continuous			
[ ] Definitely no	[ ] Probably no or unclear	[X] Probably yes	[ ] Definitely yes
Analysis based on exploratory cut point(s), e.g., picking cut point associated with highest interaction p-value	*Analysis based on cut point(s) of unclear origin*	*Analysis based on pre-specified cut point(s), e.g., suggested by prior RCT*	*Analysis based on the full continuum, e.g., assuming a linear or logarithmic relationship*
9 Optional: Are there any additional considerations that may increase or decrease credibility? (manual section 3.9) [X] not applicable			
10: How would you rate the overall credibility of the proposed effect modification?			



Box. ICEMAN credibility assessment of age as a potential effect modifier of the effect of living alone on allcause mortality.

individual trials reported planning a subgroup analysis for age, but none postulated a direction of effect.

Criteria 5: Does a test for interaction suggest that chance is an unlikely explanation of the apparent effect modification?

Whether the comparison is within or between studies, this criterion is the sine qua non of credible subgroup effect: if the *P*-value of the test for interaction is $\geq$ 0.1, one should probably not launch the ICEMAN exploration. ICEMAN's criteria are strict: a *P*-value $\leq$0.05

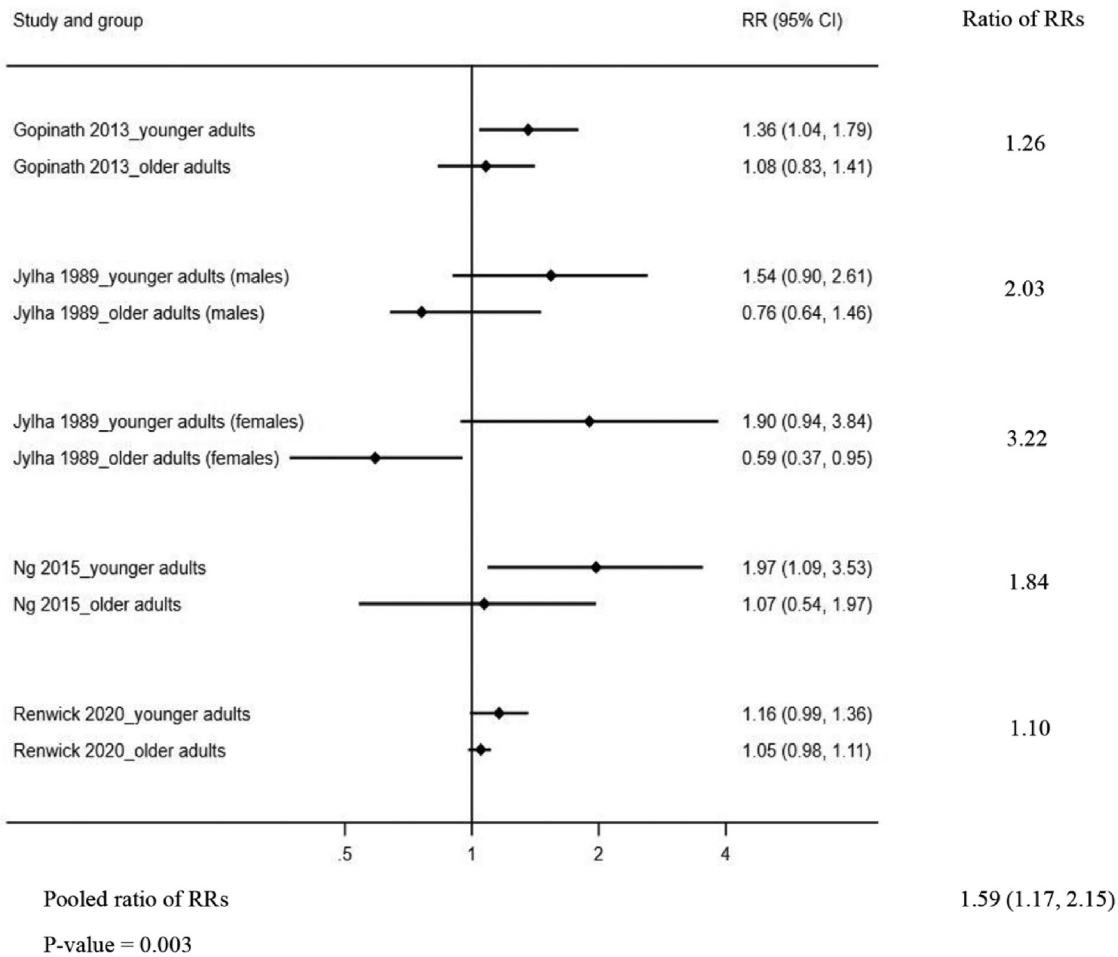


Fig. 5. Meta-analysis of within-trial subgroup analyses by age (younger vs. older adults) from five observational studies addressing the association between living alone and mortality (in: EClinicalMedicine. 2022 Sep 29; 54:101,677).

and > 0.01, meeting conventional levels of statistical significance, is considered ‘‘chance a likely explanation’’. In this case, the test of interaction (pooled ratio of relative risks 1.59, 95% CI 1.17 to 2.15, $P = 0.003$, Fig. 5) suggested that chance was an unlikely explanation for the apparent effect modification – response option ‘‘definitely yes’’.

Criteria 6: Did the authors test only a small number of effect modifiers or consider the number in their statistical analysis?

While the preferred strategy regarding testing subgroup hypotheses is to limit the number, adjustment of *P*-values for multiple comparisons provides an alternative approach. This review tested four effect modifiers (age, sex, risk of bias, and follow-up time), mandating a response of ‘‘probably no or unclear’’.

Criteria 7: Did the authors use a random effects model?

The pooled estimate of the ratio of relative risks comes from a random effects meta-analysis mandating a response option of ‘‘definitely yes’’.

Criteria 8: If the effect modifier is a continuous variable, were arbitrary cut points avoided?

One could argue the choice of thresholds in the primary studies (four at 75 and one at 65) were arbitrary. These thresholds are, however, frequently used and consistent across studies, and at the systematic review stage, authors are working with choices made by the primary study authors. The review authors therefore chose the response option of ‘‘probably yes’’.

Criterion 9: Are there any additional considerations that may increase or decrease credibility?

The systematic survey of methods papers addressing the credibility of subgroup effects identified 36 criteria offered by authors; the ICEMAN expert panel responsible for creating the instrument focused on the most important. Among the others most seriously considered [20], one possible criterion for further consideration, as part of ICEMAN’s ninth criterion addresses whether the effect is

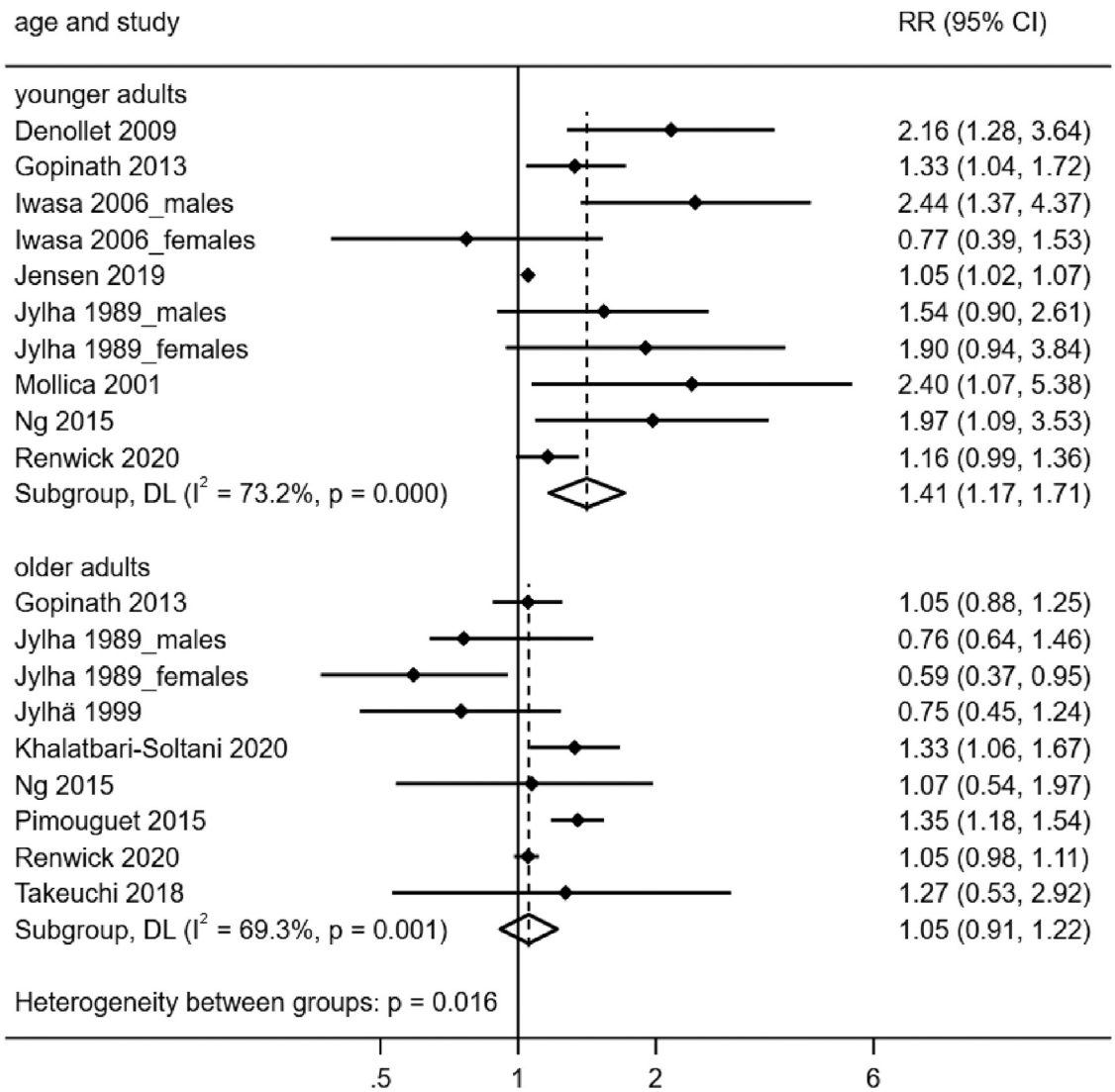


Fig. 6. Forest plot for the relative risk of adults living alone on allcause mortality according to age groups (in: EClinicalMedicine. 2022 Sep 29; 54:101,677).

independent: the panel specified that only variables measured at the time of randomization constitute candidate effect modifiers. One possible criterion for further consideration as part of ICEMAN's ninth criterion addresses whether the effect is independent: that is, if there is more than one apparent effect modifier, does the candidate modifier retain its statistical effect in a meta-regression analysis. Other possible additional criteria include the size of the subgroup effect and whether the interaction is consistent across closely related outcomes.[19] In this case, reviewers did not identify additional considerations that may increase or decrease credibility.

3.4. ICEMAN's overall rating of credibility

ICEMAN's final rating of the credibility of the subgroup effect has much in common with GRADE's rating of the certainty of evidence. First, it represents a continuum [23]; in ICEMAN's case, from very high to extremely low credibility. Second, it divides this continuum into four categories: high, moderate, low, and very low. Third, it mandates reviewers to make an overall judgment based on prior judgments of individual criteria: for evidence certainty, GRADE's five reasons for rating down and three for rating up [23]; for subgroup credibility, ICEMAN's nine individual criteria (box).

In contrast to GRADE, ICEMAN highlights the continuous nature of the credibility judgment by presenting raters with a visual analog scale, below which are the four categories of credibility (box). This allows raters to use the categorical scale while indicating where within categories their rating lies: toward the upper or lower end of high, moderate, low, or very low.

ICEMAN suggests a process for moving from the nine individual items to the overall rating: all responses definitely or probably decrease credibility, rate to very low; if two or more responses decrease credibility, rate as low or

lower; if one response definitely decreases or two responses probably reduce credibility, rate as moderate or lower; and if no responses definitely or probably decrease probability, very likely rate as high. These suggestions are, however, only a guide.

In the review of living alone as a possible influence on mortality, the authors noted concerns in two items. First, they had not specified in their protocol age as an effect modifier − although individual studies had. Second, they tested four possible effect modifiers, leading to a "probably no" for the corresponding category. However, testing four hypotheses is at the borderline of the judgment: three would lead to a "yes" if specified in the protocol and "probably no" ranges from 4 to 10 prespecified. Considering the individual judgments, the authors of the systematic review rated the credibility as at the lower end of the high credibility rating (see appendix). GRADE working group members, reviewing the same ICEMAN ratings of individual items, rated the credibility at the upper end of moderate credibility (box). This demonstrates first the inevitable subjective nature of the final overall rating; the usefulness of having both the continuum and the four categories together; and the possible situation, when ratings are near one of the thresholds, of apparent disagreement (moderate vs. high) and actual agreement (only small differences in ratings on the absolute scale).

3.5. Acting on the overall rating of credibility

ICEMAN suggests using overall estimates if the credibility of the subgroup effect is very low or low credibility (for low, noting remaining uncertainty in whether a subgroup effect exists, and therefore consider rating down for inconsistency of results across the subgroups) and, rather than the overall effect, using separate effects for high and moderate credibility (for moderate, noting remaining uncertainty).

Following this guidance, the authors of the living alone and mortality review focused on separate evidence summaries and conclusions for younger and older individuals. The issue then arose as to the sources of evidence for the point estimates and confidence intervals for each group. One possibility would be to use the estimates from only the studies that contributed to the within-study comparisons. Three studies, however, enrolled only younger individuals, and the authors reasoned that these should be combined with the younger groups from the within-study comparisons, leading to a final estimated RR of 1.41 (95% CI 1.17−1.71). Similarly, four studies enrolled only older individuals, and combining these with the older groups from the within-study comparison yielded a RR of 1.05 (95% CI 0.91−1.22) (Fig. 6).

Ultimately, authors presented the point estimates and confidence intervals and rated the certainty of the evidence regarding both the association of living alone with mortality and its possible causal impact [24], separately for older and younger individuals. They chose to rate certainty in a nonzero effect (threshold, RR = 1.0). For younger individuals, they did not identify concerns in any of risk of bias, indirectness, or inconsistency, and because the confidence interval did not cross the chosen threshold, concluded that there existed high certainty evidence that living alone is associated with increased mortality in younger individuals. Considering the issue of whether living alone actually causes reduced life span, rather than a noncausal association, because the evidence came from observational studies, authors judged the evidence to have low certainty.

Table 3. GRADE's approach to addressing subgroup analyses

1. For the PICO at hand, consider possible absolute and relative subgroup effects that may be present
2. For absolute effects:
i) Seek observational studies that define patient groups at varying baseline risk
ii) For populations in whom easily identifiable patient characteristics identify substantial differences in baseline risk, present in summary of findings tables absolute benefits for each subpopulation
iii) For guidelines, consider whether populations at varying baseline risk warrant different recommendations
3. For relative effects:
i) Generate a priori subgroup hypotheses, including postulated direction of effects, ideally based on prior evidence. Be parsimonious in choice of subgroups to investigate, ideally no more than three.
ii) When evidence available, check for possible subgroup effects, ideally focusing on within-study comparisons
iii) The check for possible subgroup effects should begin with calculation of a test of interaction. If the *P*-value for the test is 0.1 or higher, no further exploration of subgroup effect is warranted.
iv) If the *P*-value of test of interaction is smaller than 0.1, apply ICEMAN or, alternatively, one of the available checklists for determining the credibility of the subgroup analysis.
v) If the credibility is very low or low, present in summary of finding tables the overall estimate. If credibility is moderate or high, each subgroup should have its own summary of finding table with relative and absolute effects and a certainty rating.
vi) For guidelines, consider whether different recommendations are warranted for each subgroup (very likely that different recommendations will be warranted).

In older individuals,. the judgments for risk of bias, indirectness, and inconsistency identified no serious concerns, but the precision judgment presented challenges. The confidence interval crossed the threshold of no effect, but the point estimate was sufficiently near 1.0 (1.05) to suggest the possible advisability of rating certainty for little or no effect [8].

Addressing that issue requires transforming relative to absolute effects. Doing so requires choosing a baseline risk from studies of those not living alone and applying the relative effect to that baseline risk. The median follow-up in the studies of older individuals was 6.5 years, and a reasonable choice of baseline risk, approximately 19%, comes from the largest study in that group. Applying the relative risk increase of 5% to that baseline risk yields an increase in mortality of approximately 1% over 6 years (1.7% fewer deaths to 4.2% more). The authors considered a 1% increase trivial and therefore rated their certainty in little or no association or causal connection. However, they considered a 4.2% increase important, and thus rated it down for imprecision. The authors concluded that for younger individuals, the results provided moderate certainty evidence of little or no association of living alone with mortality, and very low certainty evidence regarding causality. The appendix presents the summary of findings tables, considering both association and causation, for younger and older adults.

3.6. Summary of GRADE's approach to subgroup analysis

Table 3 Summarizes GRADE's approach to subgroup analysis for both absolute and relative subgroup effects.

4. Conclusion

In this article, we have summarized insights, gained since our original guidance published in 2011, into GRADE's approach to addressing inconsistency in bodies of evidence. These include clarifying what it is in which one is rating one's certainty and emphasizing proper interpretation and use of I^2 when assessing inconsistency. Review authors should always anticipate findings of large heterogeneity, and generate a priori hypotheses, including direction, that might provide an explanation. We have focused on applying subgroup analysis to exploration of relative effects (Table 3), presenting in detail one option for rating credibility of a claimed subgroup effect, the ICEMAN instrument [20].

Acknowledgment

We would like to acknowledge Signe Flottorp for the extremely helpful summaries of feedback received at the GRADE working group meeting in July, 2022.

Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jclinepi.2023.03.003.

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